```
root@DESKTOP-4CP3D54:~# mkdir 11249A086_os_lab
mkdir: cannot create directory ‘11249A086_os_lab’: File exists
root@DESKTOP-4CP3D54:~# cd 11249A086_os_lab
root@DESKTOP-4CP3D54:~/11249A086_os_lab# pwd
/root/11249A086_os_lab
root@DESKTOP-4CP3D54:~/11249A086_os_lab# cd
root@DESKTOP-4CP3D54:~# pwd
/root
root@DESKTOP-4CP3D54:~# ls
11249A086_os_lab
root@DESKTOP-4CP3D54:~# rmdir oslab
rmdir: failed to remove 'oslab': No such file or directory
root@DESKTOP-4CP3D54:~# rmdir os
rmdir: failed to remove 'os': No such file or directory
root@DESKTOP-4CP3D54:~# cd 11249A086_os_lab
root@DESKTOP-4CP3D54:~/11249A086_os_lab# touch a.txt
root@DESKTOP-4CP3D54:~/11249A086_os_lab# cp a.txt b.txt
root@DESKTOP-4CP3D54:~/11249A086_os_lab# mv a.txt 1.txt
root@DESKTOP-4CP3D54:~/11249A086_os_lab# rm 1.txt
root@DESKTOP-4CP3D54:~/11249A086_os_lab# ls
b.txt
root@DESKTOP-4CP3D54:~/11249A086_os_lab# file b.txt
b.txt: empty
root@DESKTOP-4CP3D54:~/11249A086_os_lab# stat b.txt
  File: b.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: 8,48    Inode: 46420       Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-04-24 09:48:13.728508534 +0000
Modify: 2026-04-24 09:48:13.728508534 +0000
Change: 2026-04-24 09:48:13.728508534 +0000
 Birth: 2026-04-24 09:48:13.728508534 +0000
root@DESKTOP-4CP3D54:~/11249A086_os_lab# top
top - 09:50:11 up 8 min,  1 user,  load average: 0.09, 0.14, 0.06
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.4 sy,  0.0 ni, 99.1 id,  0.1 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   1786.2 total,   1170.2 free,    346.1 used,    343.0 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   1440.1 avail Mem
```

```
top - 09:50:11 up 8 min,  1 user,  load average: 0.09, 0.14, 0.06
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.4 sy,  0.0 ni, 99.1 id,  0.1 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   1786.2 total,   1170.2 free,    346.1 used,    343.0 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   1440.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   21668  12900   9444 S   0.3   0.7   0:02.23 systemd
      2 root      20   0    3120   2048   1920 S   0.0   0.1   0:00.01 init-systemd(Ub
      7 root      20   0    3136   2024   1920 S   0.0   0.1   0:00.00 init
     56 root      19  -1   34040  12928  12032 S   0.0   0.7   0:00.64 systemd-journal
    102 root      20   0   25240   6528   4992 S   0.0   0.4   0:00.20 systemd-udevd
    115 systemd+  20   0   21452  12672  10496 S   0.0   0.7   0:00.13 systemd-resolve
    119 systemd+  20   0   91020   7552   6784 S   0.0   0.4   0:00.10 systemd-timesyn
    153 root      20   0    4236   2560   2432 S   0.0   0.1   0:00.00 cron
    154 message+  20   0    9536   4864   4352 S   0.0   0.3   0:00.34 dbus-daemon
    161 root      20   0   17960   8192   7424 S   0.0   0.4   0:00.18 systemd-logind
    166 syslog    20   0  222508   5248   4480 S   0.0   0.3   0:00.20 rsyslogd
    168 root      20   0    3160   2048   1920 S   0.0   0.1   0:00.00 agetty
    188 root      20   0    3116   1920   1792 S   0.0   0.1   0:00.00 agetty
    196 root      20   0  107008  22388  13184 S   0.0   1.2   0:00.18 unattended-upgr
    273 root      20   0    3120    768    768 S   0.0   0.0   0:00.00 SessionLeader
    274 root      20   0    3136   1152   1024 S   0.0   0.1   0:00.04 Relay(275)
    275 root      20   0    6072   5120   3584 S   0.0   0.3   0:00.05 bash
    276 root      20   0    6668   4224   3584 S   0.0   0.2   0:00.00 login
    322 root      20   0   20304  11136   9216 S   0.0   0.6   0:00.08 systemd
    323 root      20   0   21152   3376   1664 S   0.0   0.2   0:00.00 (sd-pam)
    345 root      20   0    6072   4864   3456 S   0.0   0.3   0:00.01 bash
   2094 root      20   0    9300   5632   3456 R   0.0   0.3   0:00.01 top
```

```
  0[||                                                        2.0%] Tasks: 24, 13 thr, 0 kthr; 1 running
  1[|                                                         0.7%] Load average: 0.10 0.11 0.06
  2[|||                                                       3.3%] Uptime: 00:10:48
  3[|                                                         0.7%]
Mem[||||||||||||||||||||                                 289M/1.74G]
Swp[                                                          0K/1.00G]

Main   I/O
  PID USER      PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
    1 root       20   0 21668 12900  9444 S   0.7  0.7  0:02.88 /sbin/init
    2 root       20   0  3120  2048  1920 S   0.0  0.1  0:00.01 /init
    7 root       20   0  3136  2024  1920 S   0.0  0.1  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
    8 root       20   0  3136  2024  1920 S   0.0  0.1  0:00.00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
    9 root       20   0  3120  2048  1920 S   0.0  0.1  0:00.00 /init
   56 root       19  -1 34040 13312 12416 S   0.0  0.7  0:00.85 /usr/lib/systemd/systemd-journald
  102 root       20   0 25240  6528  4992 S   0.0  0.4  0:00.24 /usr/lib/systemd/systemd-udevd
  115 systemd-re 20   0 21452 12672 10496 S   0.0  0.7  0:00.13 /usr/lib/systemd/systemd-resolved
  119 systemd-ti 20   0 91020  7552  6784 S   0.0  0.4  0:00.11 /usr/lib/systemd/systemd-timesyncd
  142 systemd-ti 20   0 91020  7552  6784 S   0.0  0.4  0:00.00 /usr/lib/systemd/systemd-timesyncd
  153 root       20   0  4236  2560  2432 S   0.0  0.1  0:00.00 /usr/sbin/cron -f -P
  154 messagebus 20   0  9668  4864  4352 S   0.0  0.3  0:00.48 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
  161 root       20   0 17960  8192  7424 S   0.0  0.4  0:00.22 /usr/lib/systemd/systemd-logind
  166 syslog     20   0  217M  5248  4480 S   0.0  0.3  0:00.06 /usr/sbin/rsyslogd -n -iNONE
  168 root       20   0  3160  2048  1920 S   0.0  0.1  0:00.01 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
  188 root       20   0  3116  1920  1792 S   0.0  0.1  0:00.01 /sbin/agetty -o -p -- \u --noclear - linux
  189 syslog     20   0  217M  5248  4480 S   0.0  0.3  0:00.12 /usr/sbin/rsyslogd -n -iNONE
  190 syslog     20   0  217M  5248  4480 S   0.0  0.3  0:00.00 /usr/sbin/rsyslogd -n -iNONE
  191 syslog     20   0  217M  5248  4480 S   0.0  0.3  0:00.08 /usr/sbin/rsyslogd -n -iNONE
  196 root       20   0  104M 22388 13184 S   0.0  1.2  0:00.18 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
  236 root       20   0  104M 22388 13184 S   0.0  1.2  0:00.00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
  273 root       20   0  3120   768   768 S   0.0  0.0  0:00.00 /init
  274 root       20   0  3136  1152  1024 S   0.0  0.1  0:00.06 /init
  275 root       20   0  6072  5120  3584 S   0.0  0.3  0:00.06 -bash
  276 root       20   0  6668  4224  3584 S   0.0  0.2  0:00.00 /bin/login -f
  322 root       20   0 20304 11136  9216 S   0.0  0.6  0:00.08 /usr/lib/systemd/systemd --user
  323 root       20   0 21152  3376  1664 S   0.0  0.2  0:00.00 (sd-pam)
  345 root       20   0  6072  4864  3456 S   0.0  0.3  0:00.01 -bash
 2801 root       20   0  361M 20096 17408 S   0.0  1.1  0:00.07 /usr/libexec/packagekitd
 2802 root       20   0  361M 20096 17408 S   0.0  1.1  0:00.00 /usr/libexec/packagekitd
 2803 root       20   0  361M 20096 17408 S   0.0  1.1  0:00.00 /usr/libexec/packagekitd
 2804 root       20   0  361M 20096 17408 S   0.0  1.1  0:00.00 /usr/libexec/packagekitd
 2806 polkitd    20   0  300M  7552  6912 S   0.0  0.4  0:00.04 /usr/lib/polkit-1/polkitd --no-debug
Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

```
root@DESKTOP-4CP3D54:~/11249A086_os_lab# htop
root@DESKTOP-4CP3D54:~/11249A086_os_lab# kill 9
root@DESKTOP-4CP3D54:~/11249A086_os_lab# pkill root
root@DESKTOP-4CP3D54:~/11249A086_os_lab# jobs
root@DESKTOP-4CP3D54:~/11249A086_os_lab# fg
-bash: fg: current: no such job
root@DESKTOP-4CP3D54:~/11249A086_os_lab# bg
-bash: bg: current: no such job
root@DESKTOP-4CP3D54:~/11249A086_os_lab# uptime
 09:56:34 up 14 min,  1 user,  load average: 0.04, 0.08, 0.05
root@DESKTOP-4CP3D54:~/11249A086_os_lab# who
root     pts/1        2026-04-24 09:44
root@DESKTOP-4CP3D54:~/11249A086_os_lab# whoaim
Command 'whoaim' not found, did you mean:
  command 'whoami' from deb coreutils (9.4-2ubuntu2)
Try: apt install <deb name>
root@DESKTOP-4CP3D54:~/11249A086_os_lab# free
              total        used        free      shared  buff/cache   available
Mem:        1829108      401604     1366404        3556      137240     1427504
Swap:       1048576           0     1048576
root@DESKTOP-4CP3D54:~/11249A086_os_lab# df
Filesystem      1K-blocks     Used  Available Use% Mounted on
none               914552        0     914552   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none               914552        4     914548   1% /mnt/wsl
drivers         248938492 95399516  153538976  39% /usr/lib/wsl/drivers
/dev/sdd       1055762868  1213872 1000845524   1% /
none               914552       76     914476   1% /mnt/wslg
none               914552        0     914552   0% /usr/lib/wsl/lib
rootfs             909512     2720     906792   1% /init
none               914552      492     914060   1% /run
none               914552        0     914552   0% /run/lock
none               914552        0     914552   0% /run/shm
none               914552       76     914476   1% /mnt/wslg/versions.txt
none               914552       76     914476   1% /mnt/wslg/doc
C:\             248938492 95399516  153538976  39% /mnt/c
D:\             976629756 70405664  906224092   8% /mnt/d
tmpfs              182908       20     182888   1% /run/user/0
root@DESKTOP-4CP3D54:~/11249A086_os_lab#
```

```
root@DESKTOP-4CP3D54:~/11249a086_os_lab# du
4       .
root@DESKTOP-4CP3D54:~/11249a086_os_lab# cat >b.txt
hi
i am dileep
welcome to os lab
^C
root@DESKTOP-4CP3D54:~/11249a086_os_lab# cat b.txt
hi
i am dileep
welcome to os lab
root@DESKTOP-4CP3D54:~/11249a086_os_lab# tac b.txt
welcome to os lab
i am dileep
hi
root@DESKTOP-4CP3D54:~/11249a086_os_lab# more b.txt
hi
i am dileep
welcome to os lab
root@DESKTOP-4CP3D54:~/11249a086_os_lab# less b.txt
root@DESKTOP-4CP3D54:~/11249a086_os_lab# head b.txt
hi
i am dileep
welcome to os lab
root@DESKTOP-4CP3D54:~/11249a086_os_lab# tail b.txt
hi
i am dileep
welcome to os lab
root@DESKTOP-4CP3D54:~/11249a086_os_lab# grep "dileep" b.txt
i am dileep
root@DESKTOP-4CP3D54:~/11249a086_os_lab# wc b.txt
 3  8 33 b.txt
root@DESKTOP-4CP3D54:~/11249a086_os_lab# sort b.txt
hi
i am dileep
welcome to os lab
```

```
root@DESKTOP-4CP3D54:~/11249a086_os_lab# uniq b.txt
hi
i am dileep
welcome to os lab
root@DESKTOP-4CP3D54:~/11249a086_os_lab# cut -d':' -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
systemd-timesync
dhcpcd
messagebus
syslog
systemd-resolve
uuidd
landscape
polkitd
```

```
root@DESKTOP-4CP3D54:~/11249a086_os_lab# touch a.txt
root@DESKTOP-4CP3D54:~/11249a086_os_lab# paste b.txt a.txt
hi
i am dileep
welcome to os lab
root@DESKTOP-4CP3D54:~/11249a086_os_lab# tr a-z A-Z<b.txt
HI
I AM DILEEP
WELCOME TO OS LAB
root@DESKTOP-4CP3D54:~/11249a086_os_lab# vi 3.py
root@DESKTOP-4CP3D54:~/11249a086_os_lab# python3 3.py
  File "/root/11249a086_os_lab/3.py", line 3
    iam dileep
        ^^^^^^
SyntaxError: invalid syntax
root@DESKTOP-4CP3D54:~/11249a086_os_lab# nano b.txt
root@DESKTOP-4CP3D54:~/11249a086_os_lab# ls -l
total 8
-rw-r--r-- 1 root root 50 Apr 24 10:12 3.py
-rw-r--r-- 1 root root  0 Apr 24 10:10 a.txt
-rw-r--r-- 1 root root 33 Apr 24 10:13 b.txt
root@DESKTOP-4CP3D54:~/11249a086_os_lab# touch 6.sh
root@DESKTOP-4CP3D54:~/11249a086_os_lab# chmod u+x 6.sh
root@DESKTOP-4CP3D54:~/11249a086_os_lab# chown dileep 2.txt
chown: invalid user: ‘dileep’
root@DESKTOP-4CP3D54:~/11249a086_os_lab# chown dileep a.txt
chown: invalid user: ‘dileep’
root@DESKTOP-4CP3D54:~/11249a086_os_lab#
```